NETWORK BASICS 2:
HOMOPHILY & COMMUNITIES

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Common network tasks

- Node tasks
  - Node ranking/centrality
    - Identify important nodes in a network
  - Node classification
    - predict the labels (or categories) of nodes in a graph based on their features and the network structure
    - E.g., predict political affiliation of Twitter users; Identify fraudulent accounts in financial transaction networks.
- Link prediction
  - predict missing or future connections (edges) between nodes
  - E.g., future collaborations, recommend new people to follow, product recommendation in bipartite user-product networks
- Community detection
  - identify densely connected groups of nodes isolated from the rest of the network.
    - E.g., social circles, topic clusters in information networks

NODE CLASSIFICATION

Node classification methods

Predict labels for nodes based on a combination of node features and the graph's structure

- **Label Propagation**
  - semi-supervised learning algorithm that spreads known labels across a graph using the homophily assumption (users tend to interact with like-minded individuals).
- **Machine learning**
  - Train a classifier on extracted node features (e.g., degree, clustering coefficient).
- **Graph embedding**
  - Learn a low-dimensional representation (embedding) of each node that captures its position in the network and its local neighborhood structure
  - **DeepWalk** and **Node2Vec**: Use random walks on the graph to capture node relationships in a continuous vector space.

Classifying user political affiliation in COVID-19 discussions

The dataset consists of **9.8 million retweets** collected from **1.9 million Twitter users** discussing **COVID-19 pandemic** early stage of the pandemic.

- **Label Propagation Algorithm (LPA)** was applied to the full **1.9 million-user retweet network** to infer political and science-related polarization.
- The **content-based models (LDA and FastText)** were applied to users who had enough textual data available.
- The **FastText classification method** was trained on a **subset of ~900,000 users** who had enough tweets.

Rao et al. (2021). Political partisanship and antiscience attitudes in online discussions about COVID-19: Twitter content analysis. *Journal of medical Internet research*, *23*(6), e26692.

Political affiliation classification

Method	Political Polarization (Accuracy %)	Science Polarization (Accuracy %)	Moderacy (Accuracy %)
Label Propagation Algorithm (LPA)	92.3%	92.6%	20.1% (Poor Performance)
LDA (Topic Modeling)	93.5%	92.4%	86.4%
FastText (Text Embeddings)	**95.1%**	**93.8%**	**90.2%**

Classify social media users based on their ideological alignment across three dimensions:

1. **Political Ideology** (Liberal vs. Conservative)
2. **Science Attitude** (Pro-science vs. Anti-science)
3. **Moderacy** (Hardline vs. Moderate)

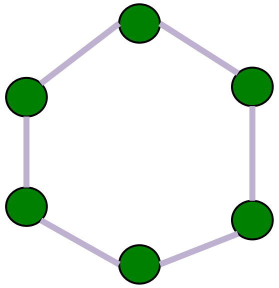
Rao et al. (2021). Political partisanship and antiscience attitudes in online discussions about COVID-19: Twitter content analysis. *Journal of medical Internet research*, *23*(6), e26692.

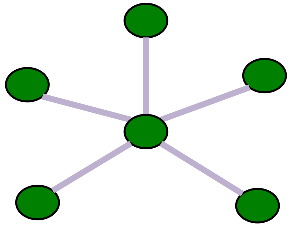
NODE CENTRALITY & RANKING

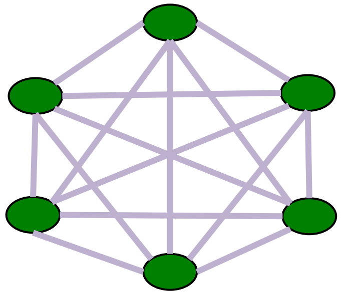
Review: How do we measure importance?

Certain positions within the social network give nodes more power or importance

- Who can directly affect/influence others?
  - Highest *degree* nodes are "in the thick of it"
- Who controls the flow of information
  - Nodes that fall on shortest paths *between* others can disrupt the flow of information between them
- Who can quickly inform most others?
  - Nodes who are *close* to other nodes can quickly get information to them
- Who has most power?
  - Nodes who are connected to other *important* nodes







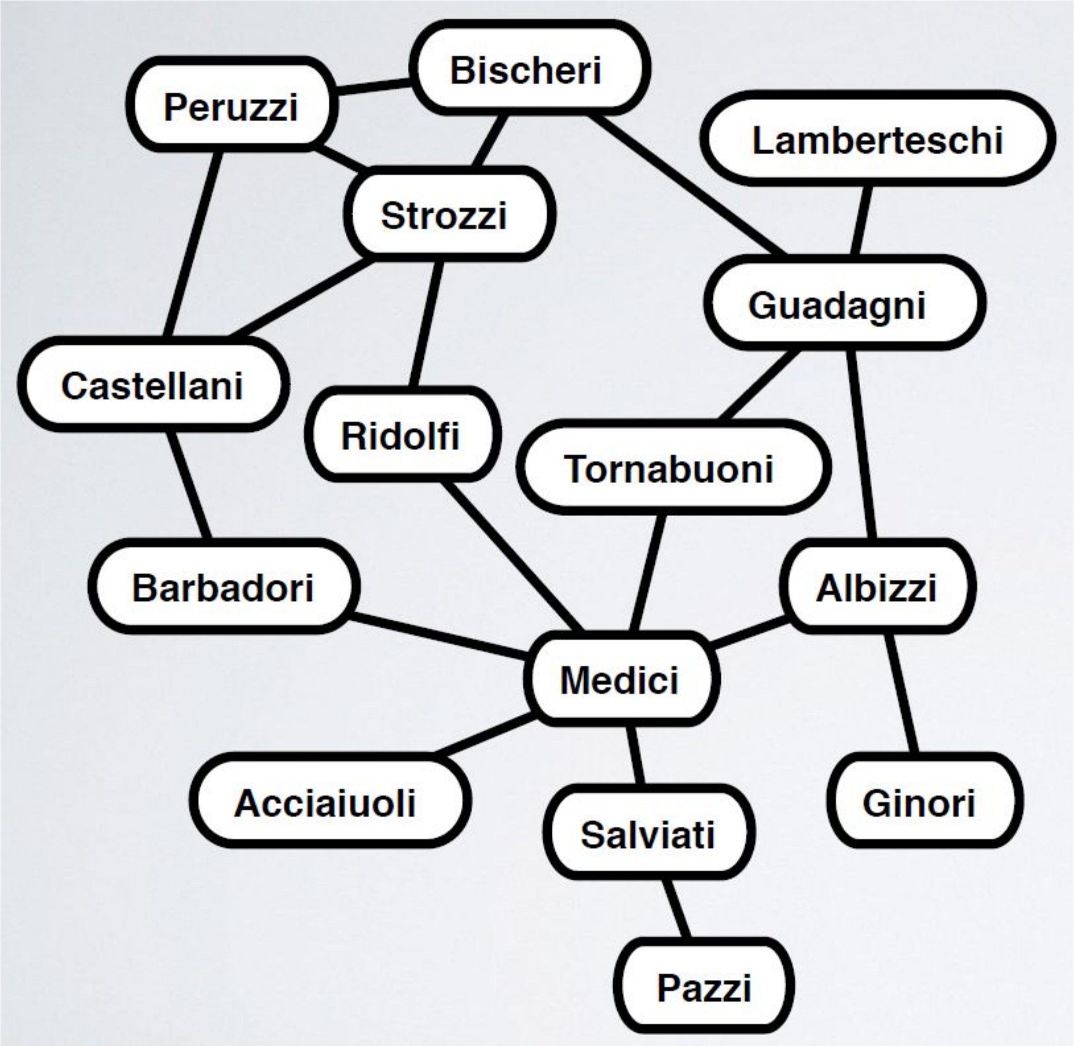
A historical perspective

Robust Action and the Rise of the Medici, 1400–1434[1]

John F. Padgett and Christopher K. Ansell

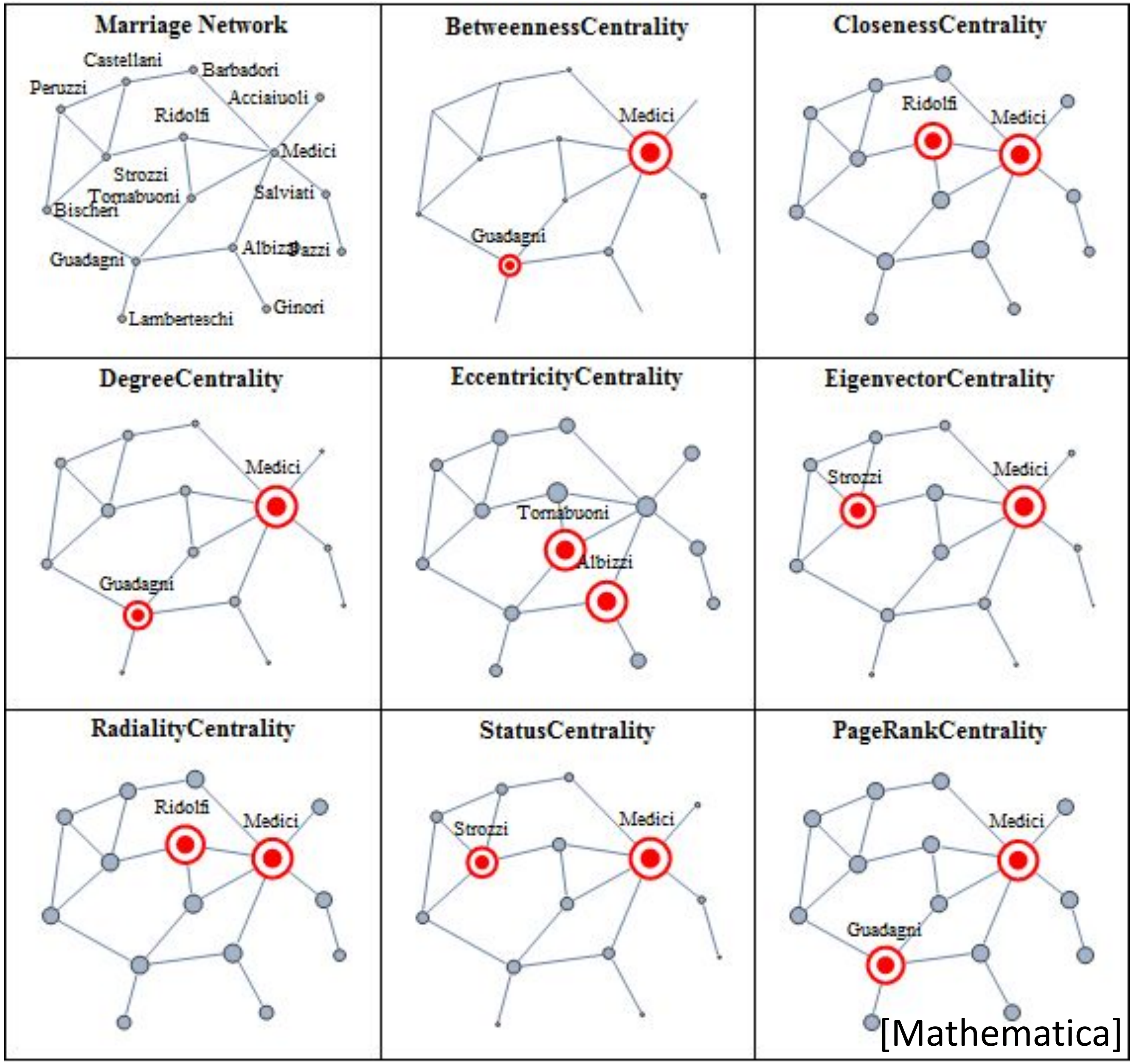
Cosimo de' Medici
1389-1464

Florentine families' marriage network



- Medici centralized control through the "star" or "spoke" network.
  - Information, social interactions passed through Medicis
- The oligarchs were densely interconnected
- In 1433, during a military showdown with the oligarchs, Medici quickly mobilized supporters, and prevented oligarchs from taking power.

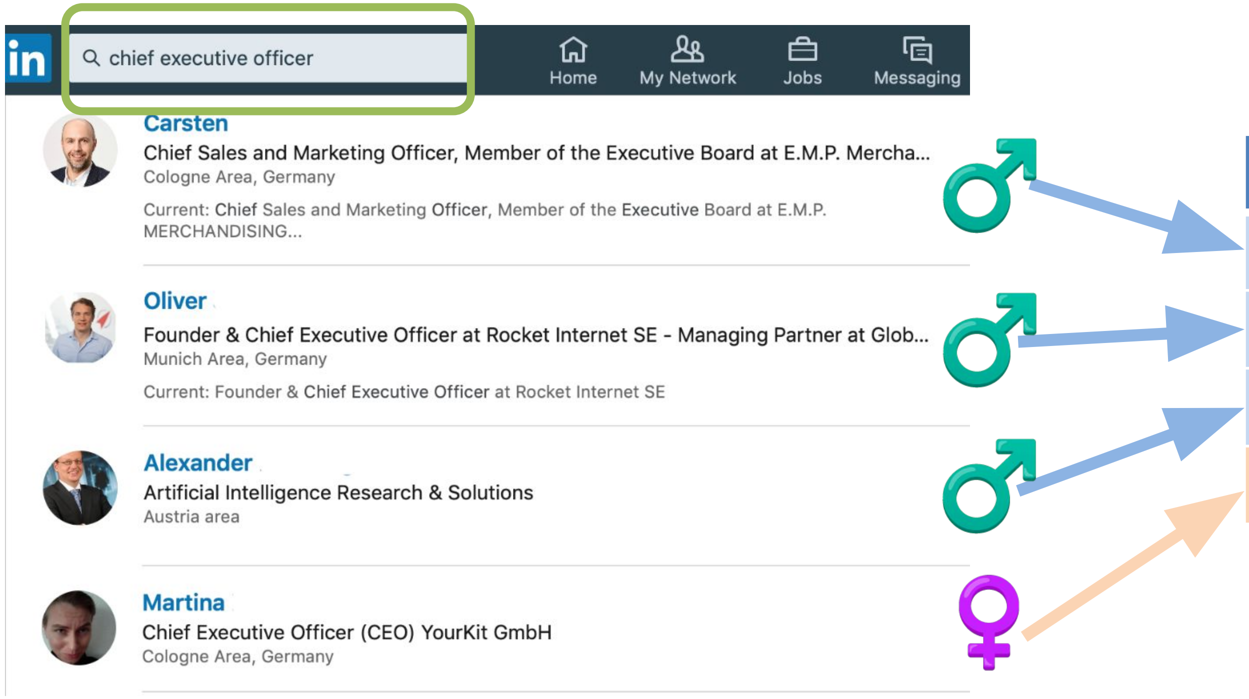
Centrality measures



Ranking people in online social networks

- Is the ranking algorithm fair?

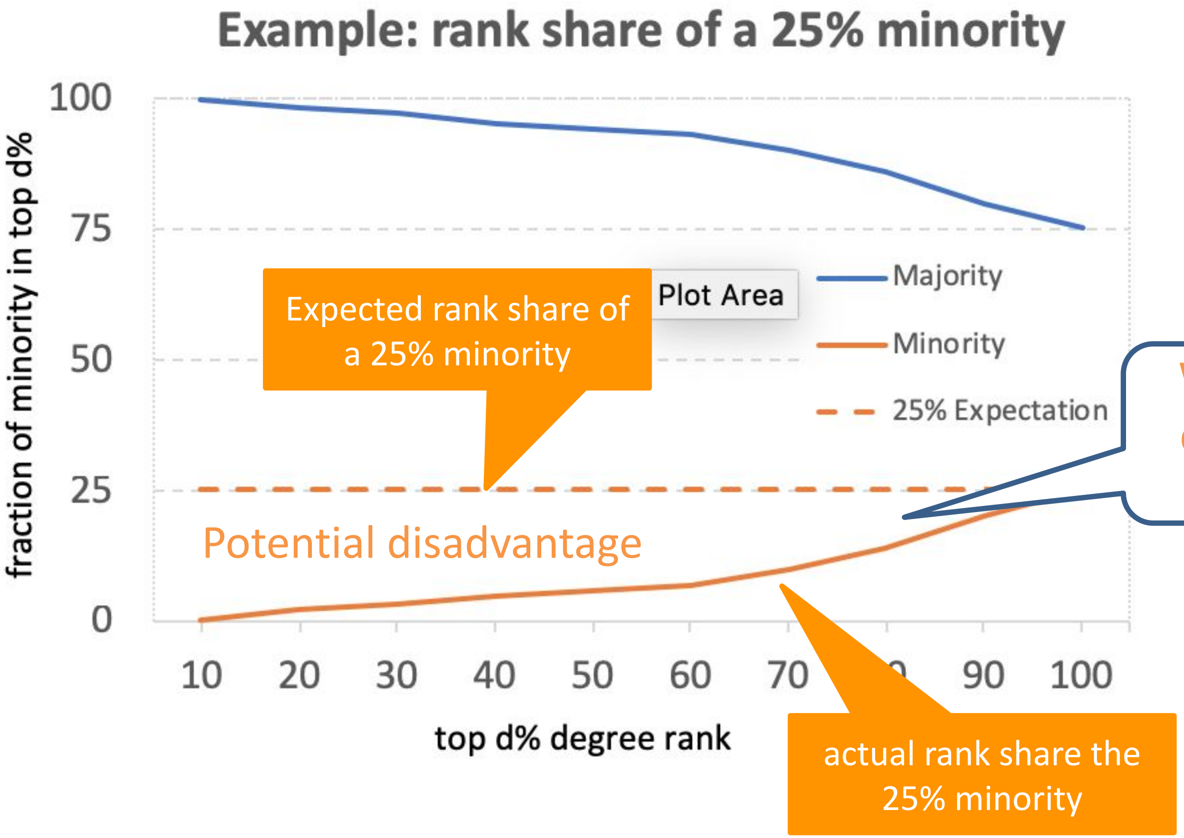
1.
2.
3.
4.

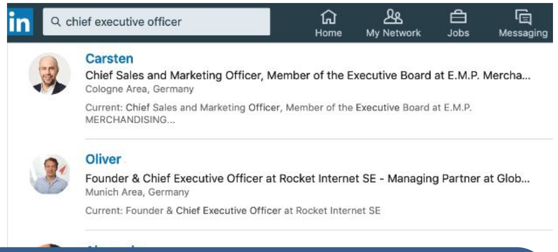


Rank	Gender
1.	m
2.	m
3.	m
4.	f

¼ (25%) women in the top 4 results

How visible are minorities in social networks



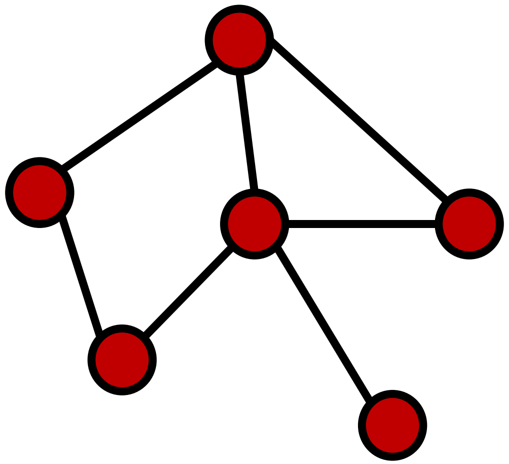


Where does this potential disadvantage come from? Algorithmic or social?

Slide courtesy of Markus Strohmaier

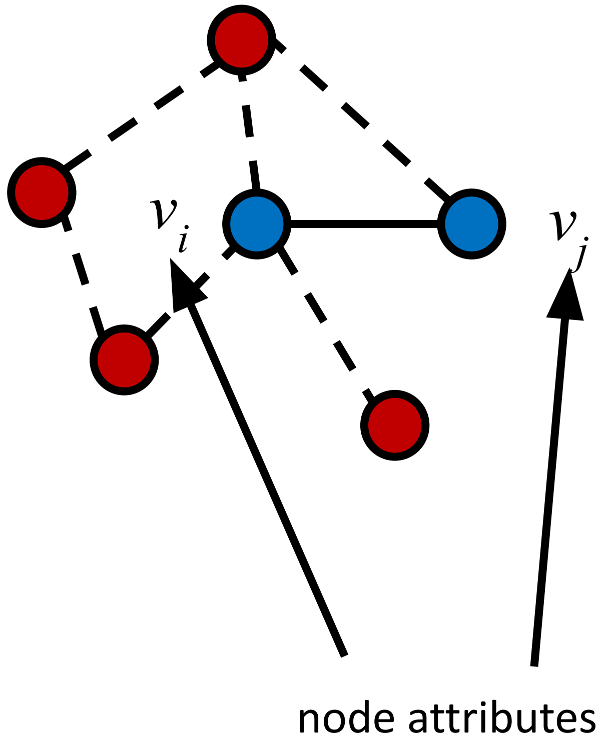
HOMOPHILY & ASSORTATIVE MIXING

Homophily and assortative mixing



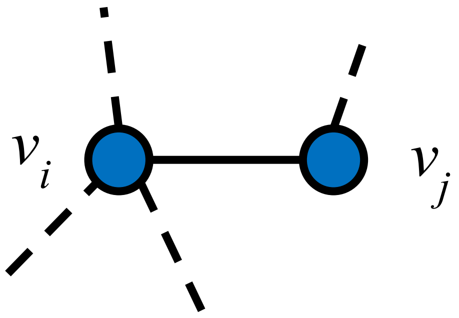
- Homophily
  - "like links to like"

Homophily and assortative mixing

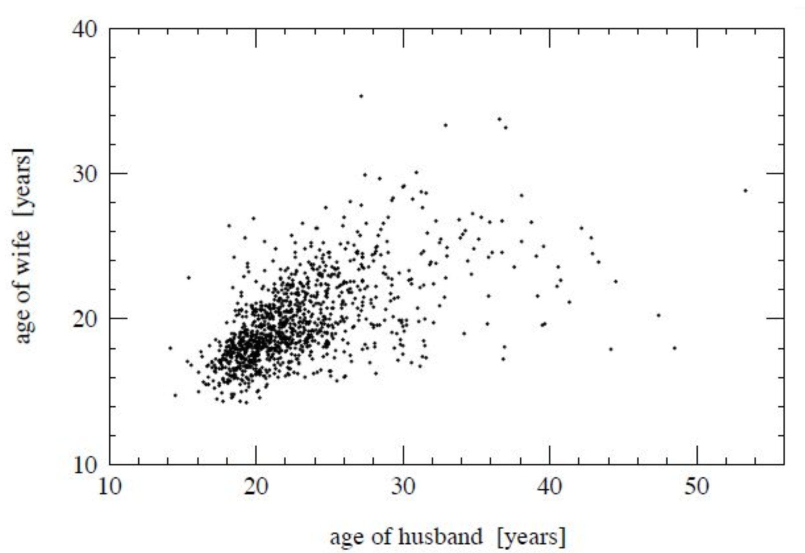


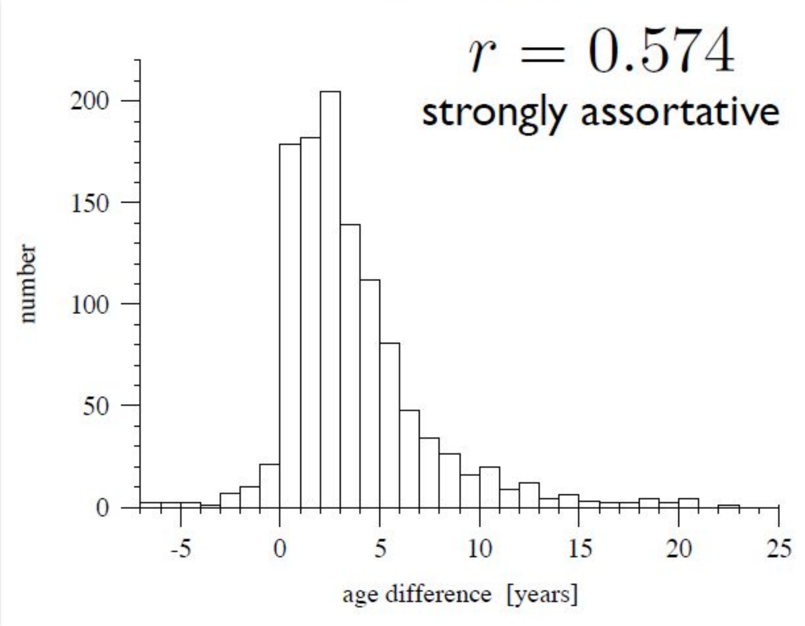
- Homophily
  - “like links to like”
- Assortativity coefficient *r* quantifies homophily
  - Pearson correlation across edges
  - -1 <= *r* <= 1
- three types:
  - scalar attributes
  - node degrees
  - categorical variables

Assortative mixing (homophily)

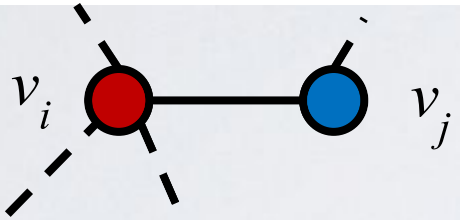


- (top) scatter plot of ages of 1141 married couples at time of marriage [1995 US National Survey of Family Growth]
- (bottom) histogram of age differences (M-F) for same data





Assortative mixing (homophily)



		women				
		black	hispanic	white	other	a_i
men	black	0.258	0.016	0.035	0.013	0.323
	hispanic	0.012	0.157	0.058	0.019	0.247
	white	0.013	0.023	0.306	0.035	0.377
	other	0.005	0.007	0.024	0.016	0.053
	b_i	0.289	0.204	0.423	0.084	

1992 study of heterosexual partnerships in San Francisco

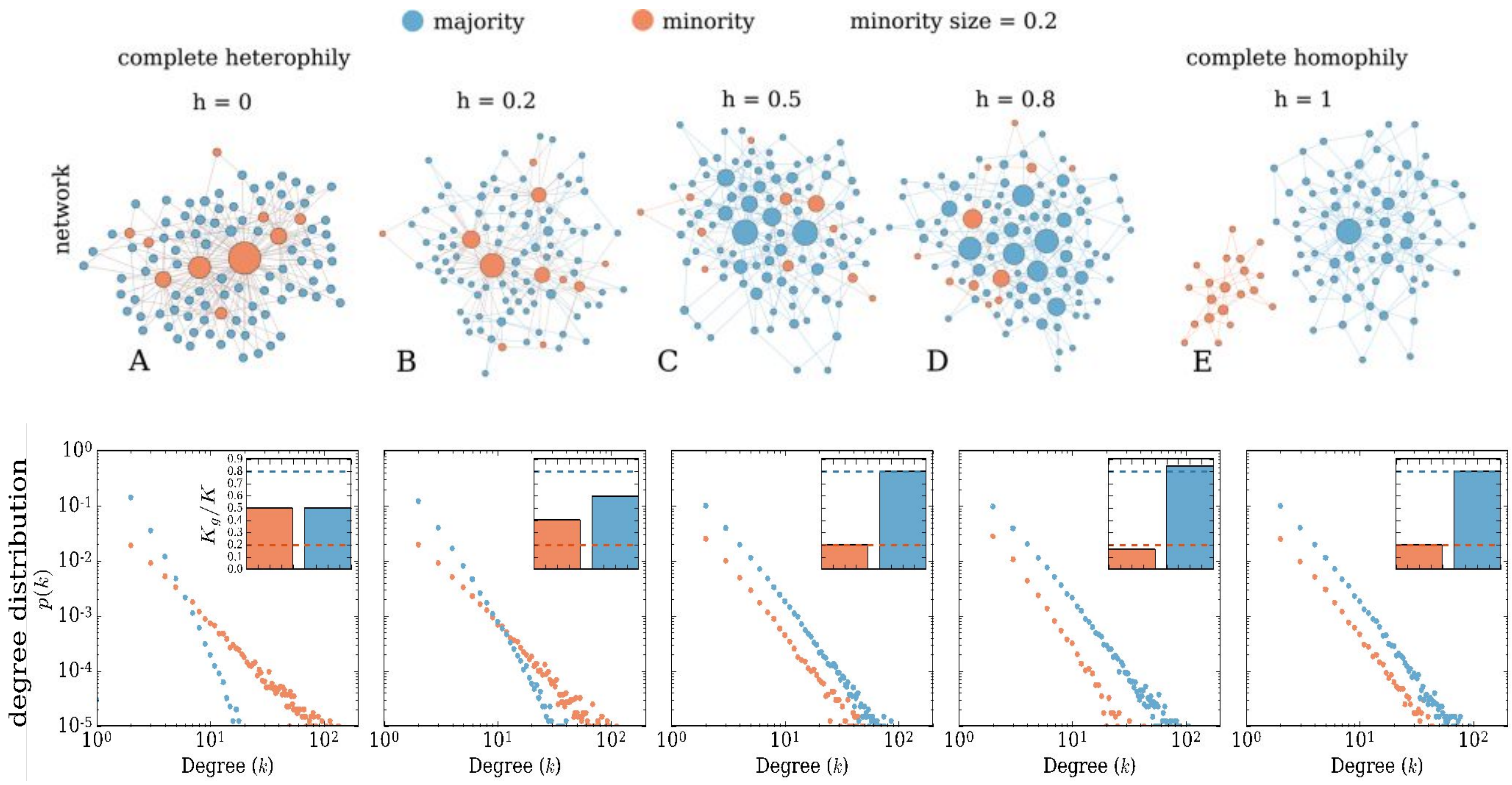
Catania et al., *Am. J. Public Health* **82**, 284-287 (1992).

Homophily influences ranking of minorities in social networks

Fariba Karimi, Mathieu Génois, Claudia Wagner, Philipp Singer & Markus Strohmaier

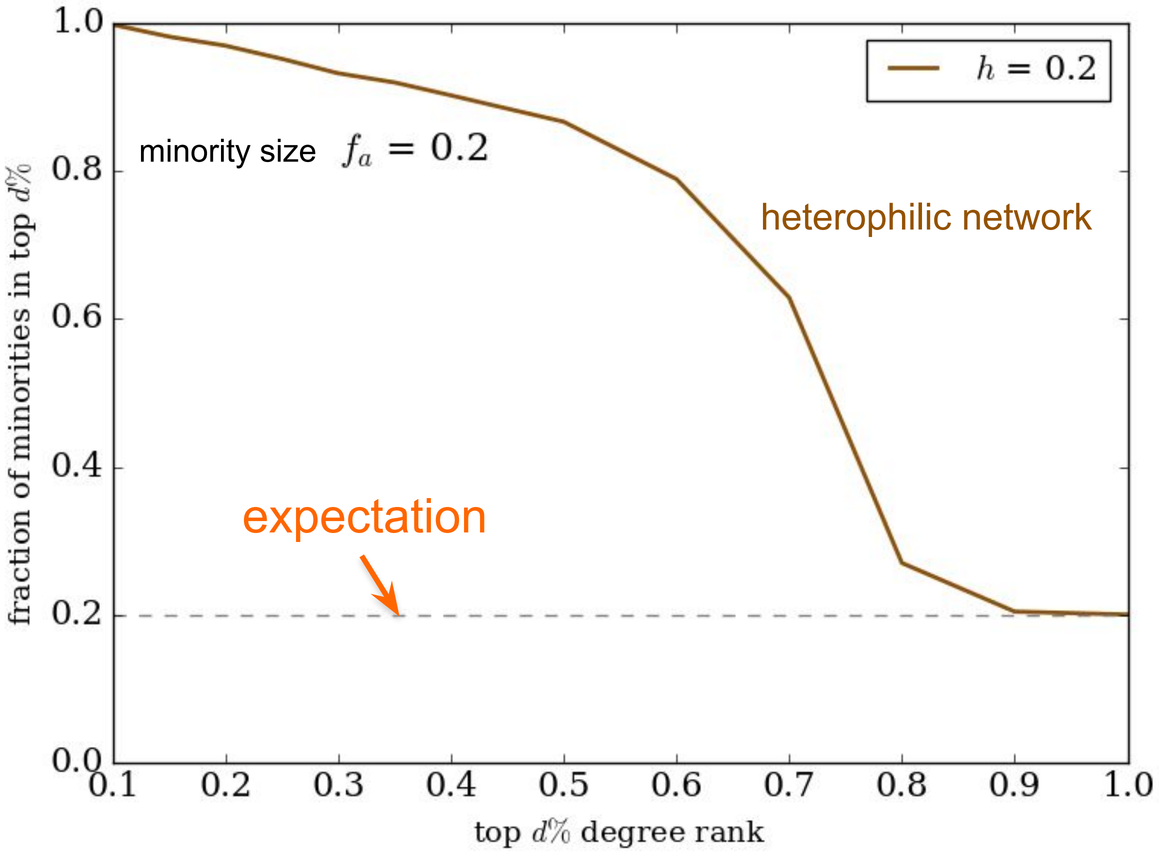
homophily and group size differences affect the structure of a social network and ranking of minorities

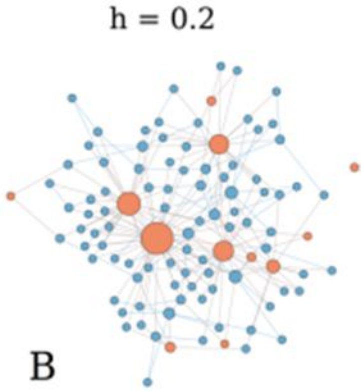
How does homophily influence degree distributions of minorities?



Ranking of minorities in social networks

- Visibility of minority in top d%





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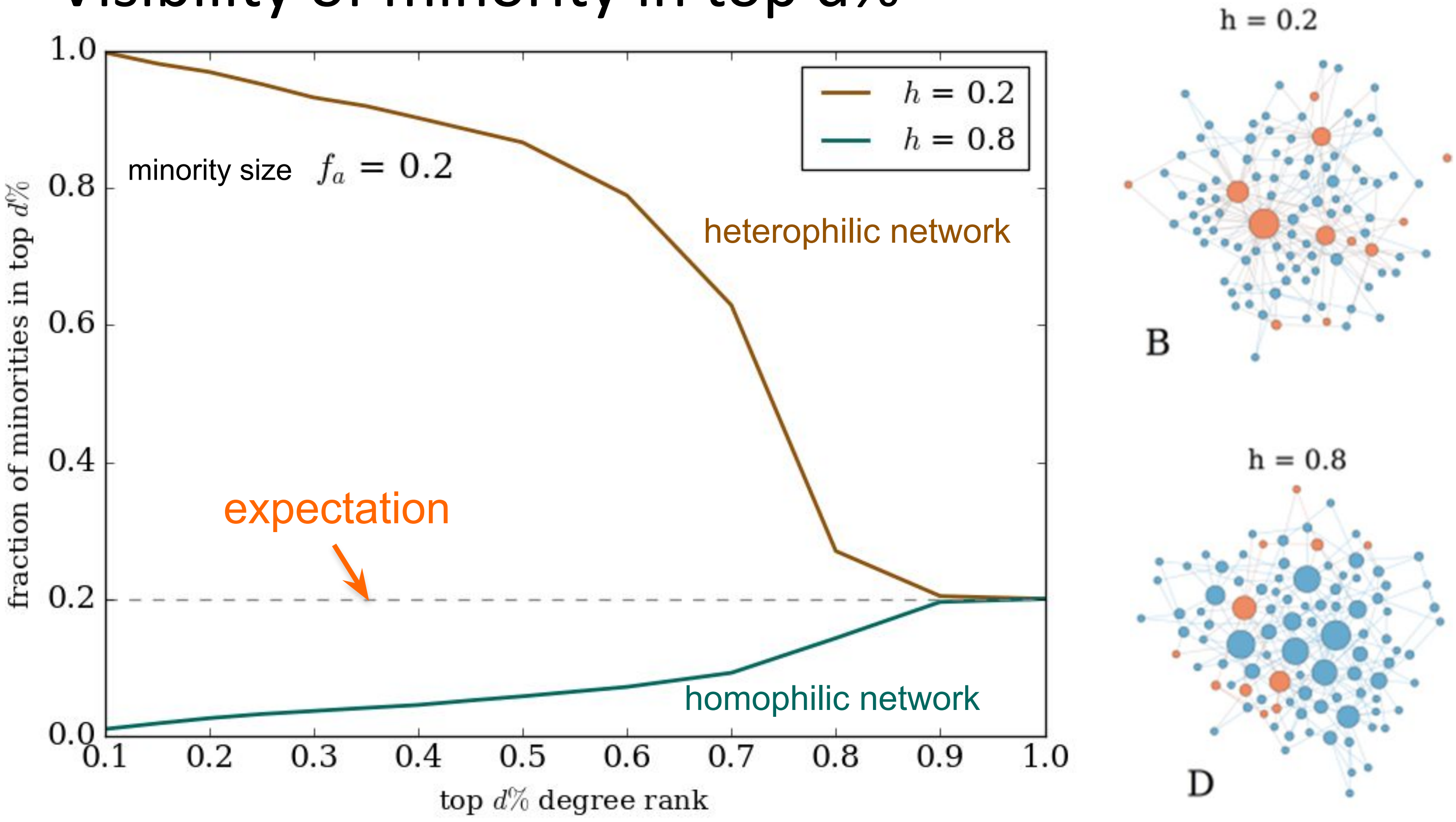

Ranking of minorities in social networks

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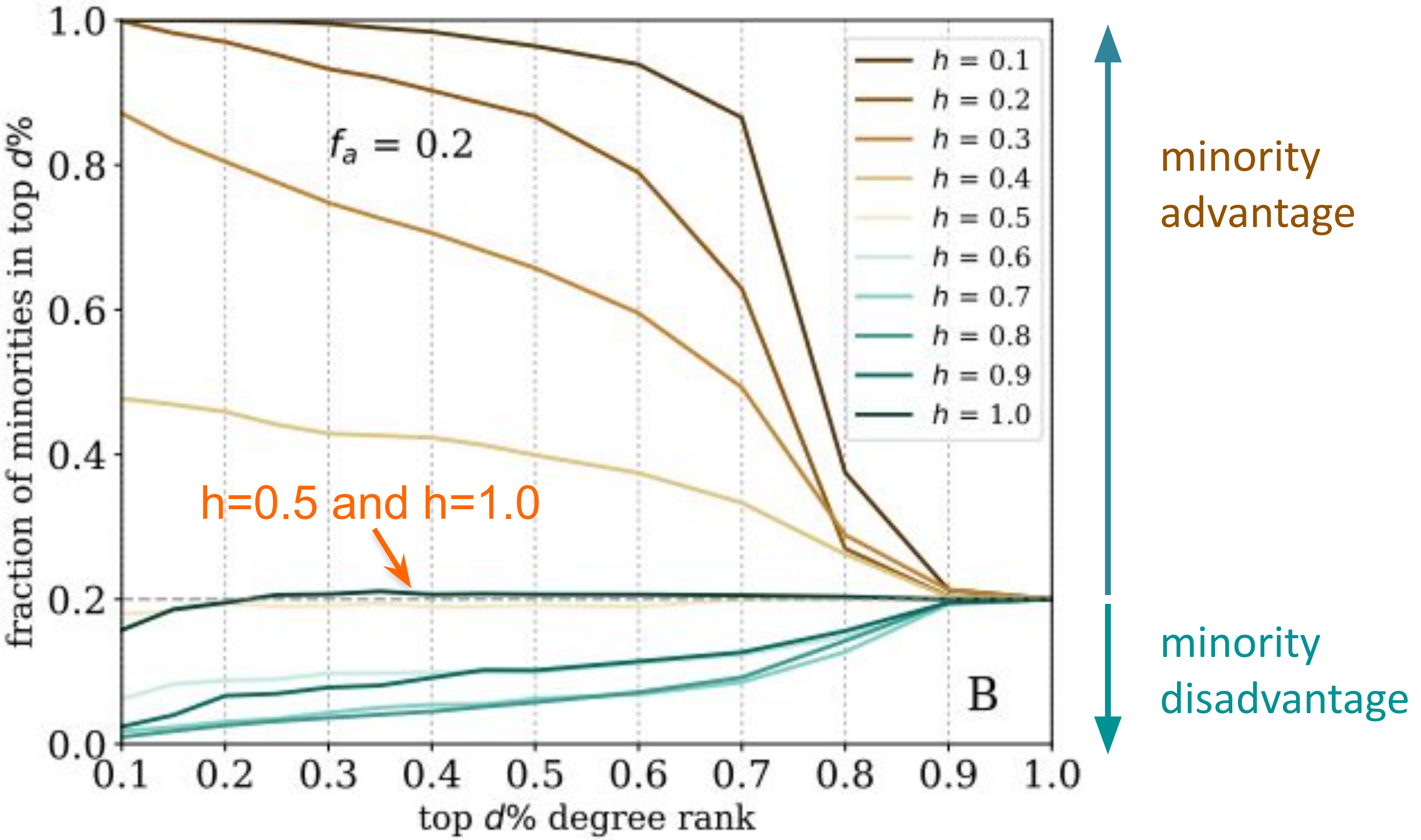
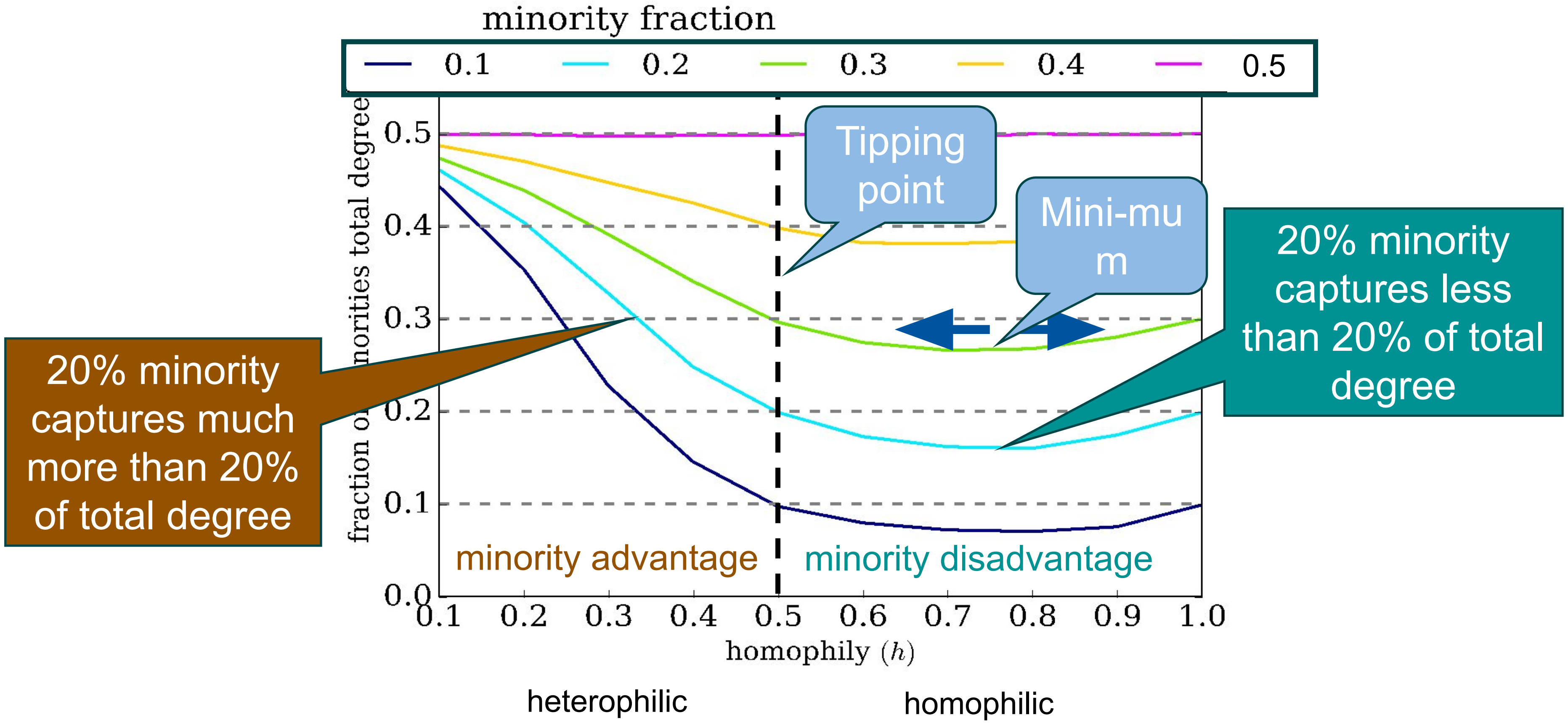


Ranking of minorities in social networks

- Fraction of minorities total degree vs. homophily



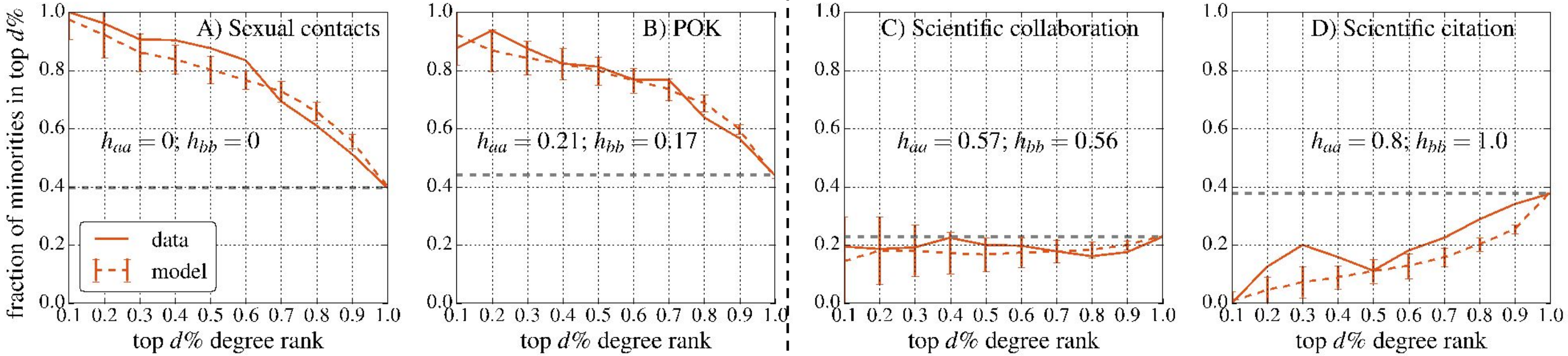
Inequalities in ranking of minorities

Heterophilic networks

Homophilic networks

minority advantage

minority disadvantage



Online prostitution	Online dating	Co-author network Comp. Science	Scientific citation Academic literature
Minority: Sex workers	Minority: Women	Minority: Women	Minority: Scientific subfield

COMMUNITY DETECTION

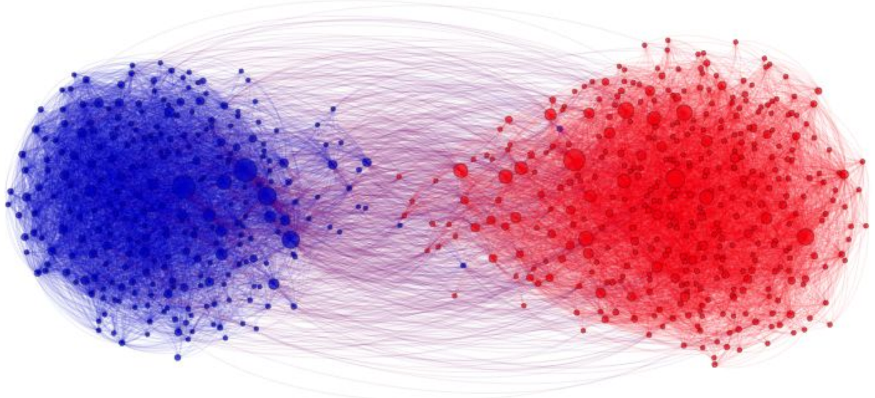
Assortativity creates communities

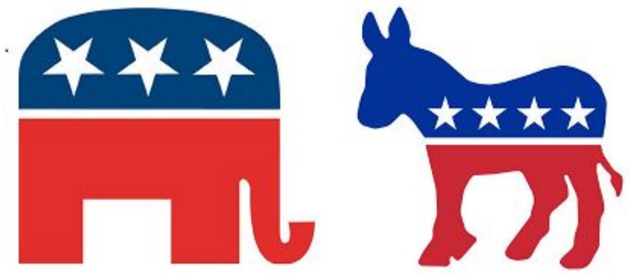
Analyzing communities helps better understand users and social processes

– Users form groups based on their interests

Groups provide a clear global view of user interactions

- E.g., find polarization





Some behaviors are only observable in a group setting and not on an individual level

– Some republican can **agree** with some democrats, but their parties can **disagree**

Social Media Communities

- **Formation:**
  - When like-minded users on social media form a link and start interacting with each other
- **More Formal Formation**:
  1. A set of at least two nodes sharing some interest, and
  2. Interactions with respect to that interest.

- Social Media Communities
  - **Explicit** (*emic*): formed by user subscriptions
  - **Implicit** (*etic*): implicitly formed by social interactions
    - **Example:** individuals calling Canada from the United States
    - Phone operator considers them one community for promotional offers
- Other community names: *group*, *cluster*, *cohesive subgroup*, or *module*

Examples of Explicit Social Media Communities

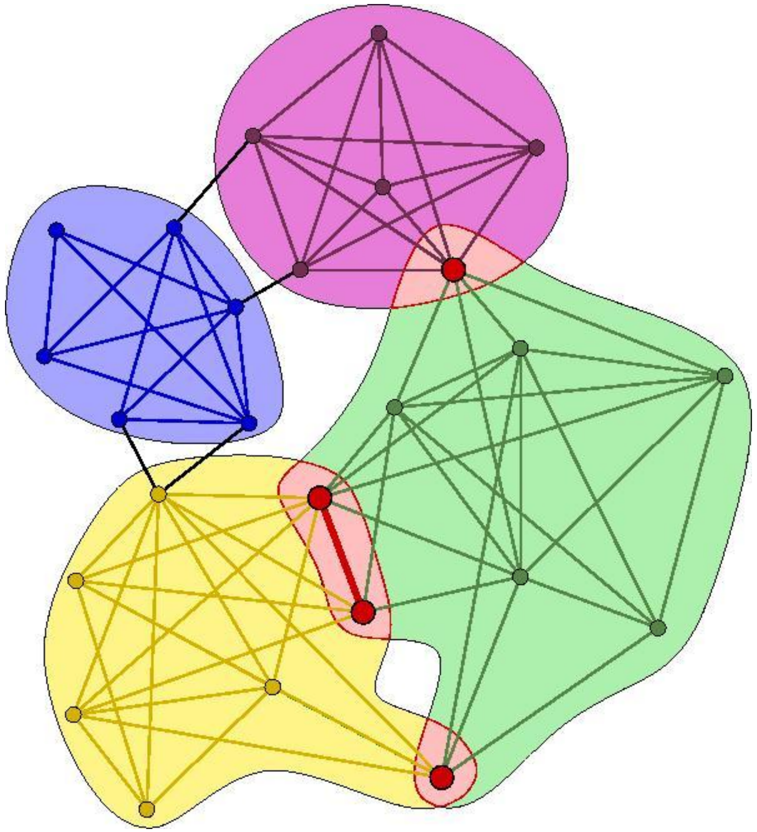
Facebook has groups and communities. Users can

- post messages and images,
- can comment on other messages,
- can like posts, and
- can view activities of other users

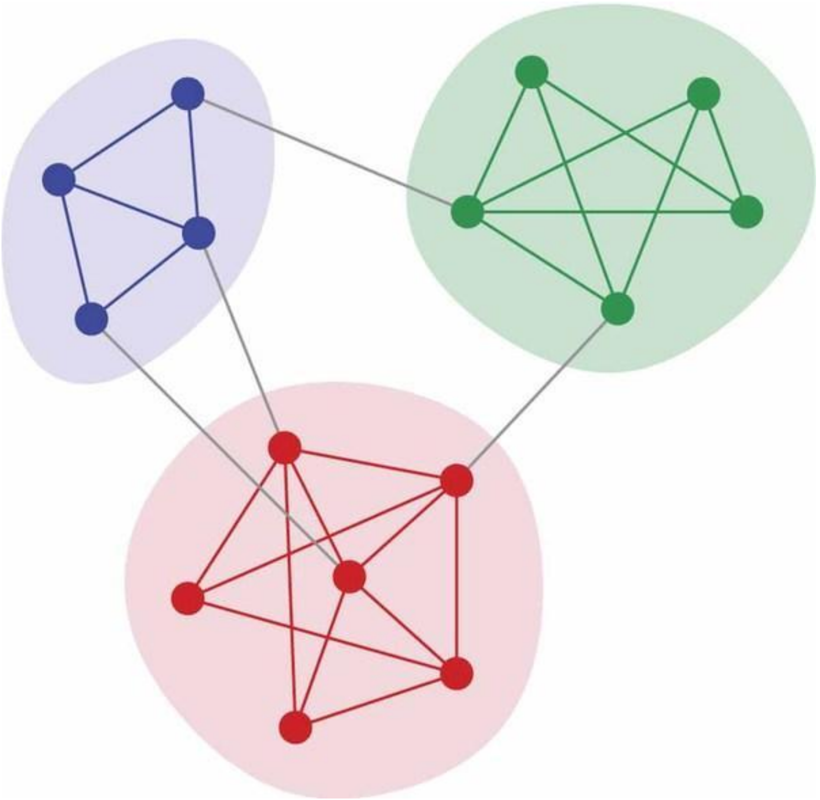
LinkedIn provides *Groups* and *Associations*.

- Users can join professional groups where they can post and share information related to the group

Overlapping vs. Disjoint Communities

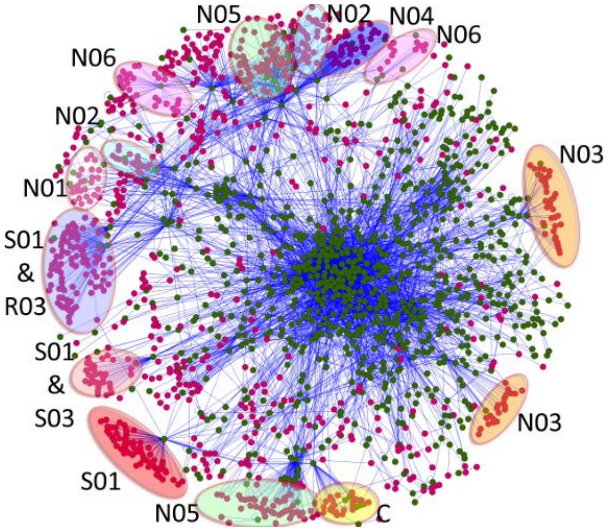


Overlapping Communities



Disjoint Communities

Implicit communities in other domains



Protein-protein interaction networks

– Communities are likely to group proteins having the same specific function within the cell

World Wide Web

– Communities may correspond to groups of pages dealing with the same or related topics

Metabolic networks

– Communities may be related to functional modules such as cycles and pathways

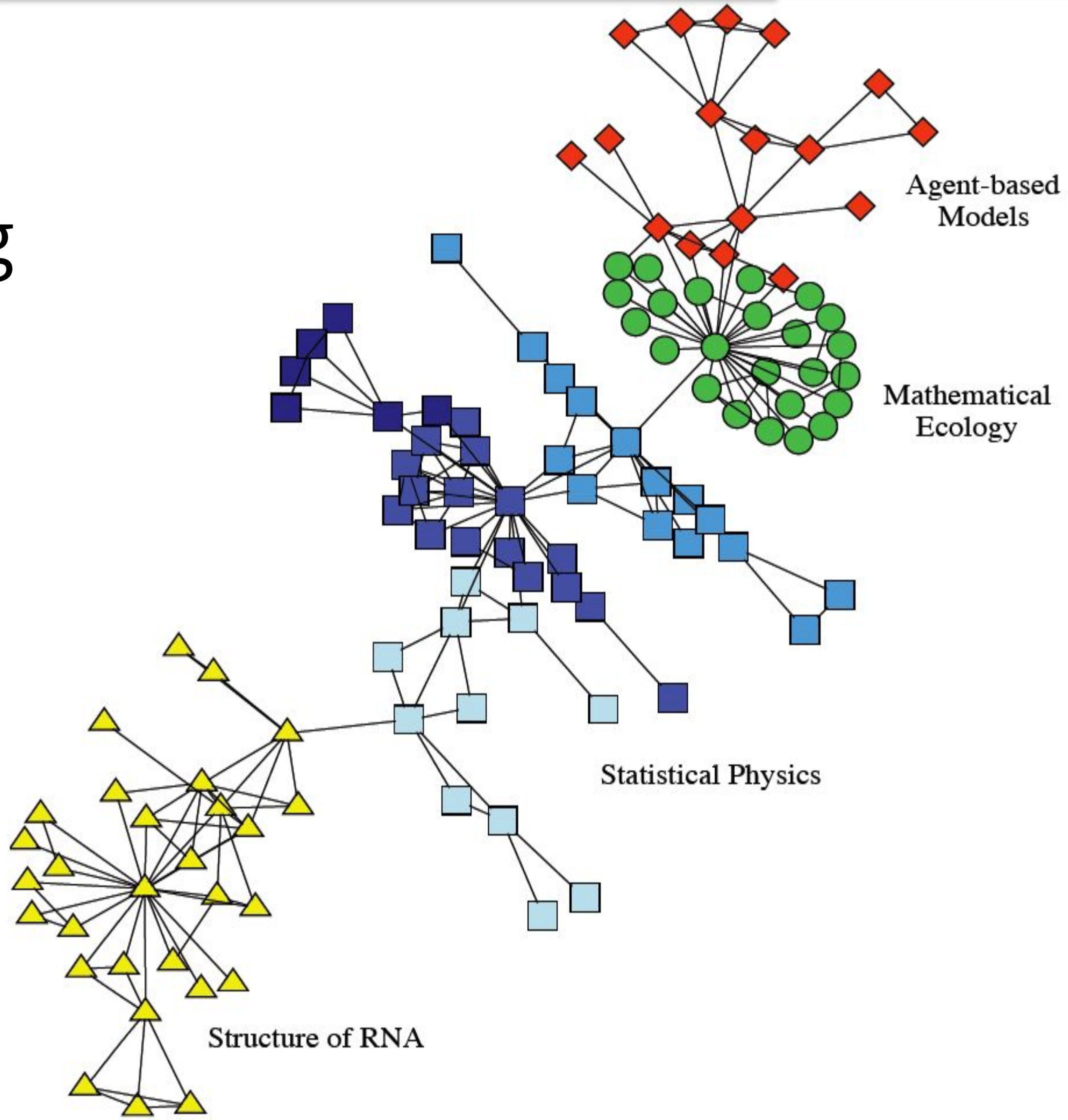
Food webs

– Communities may identify compartments

Real-world Implicit Communities

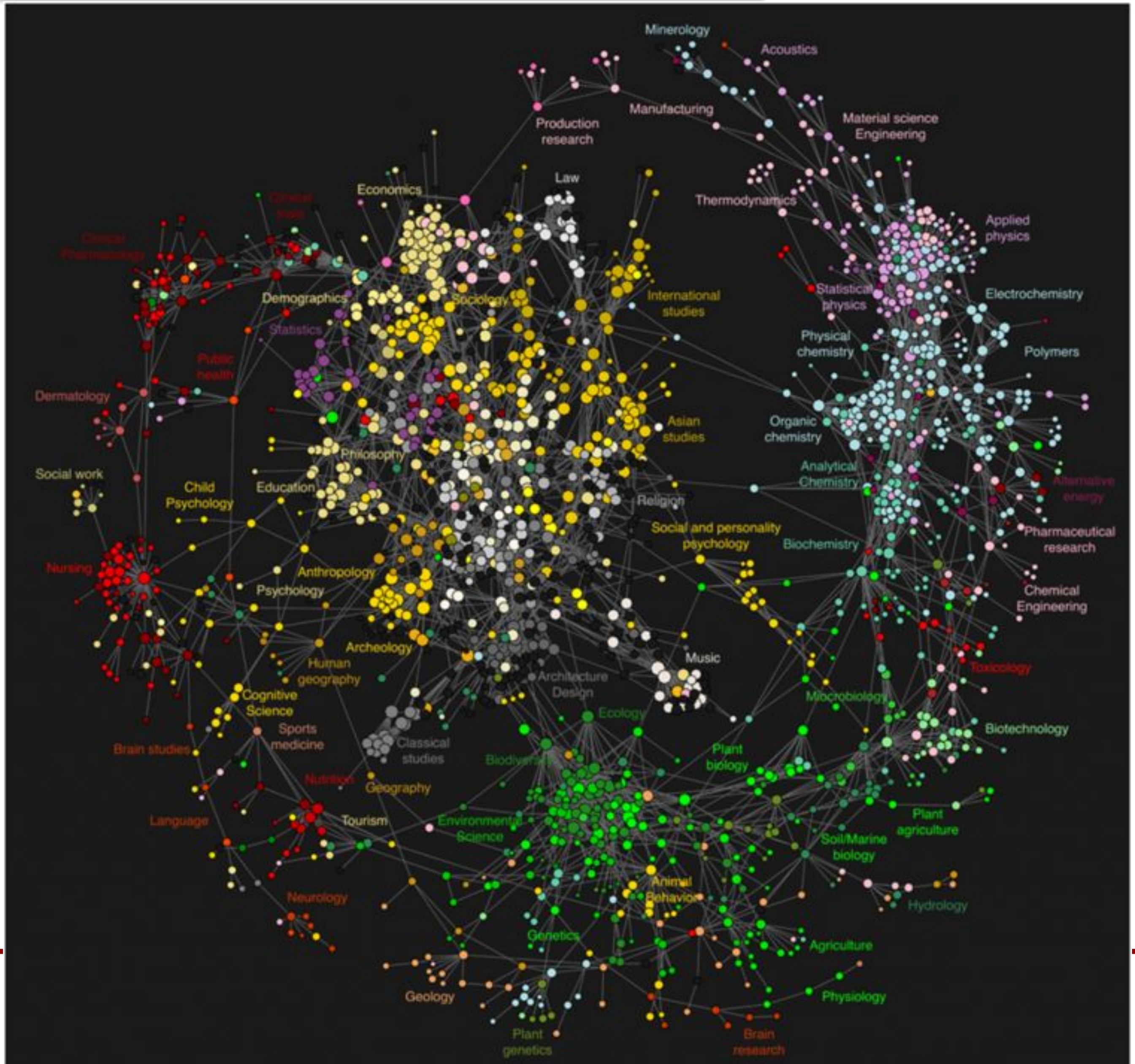
Collaboration network between scientists working at the Santa Fe Institute.

The colors indicate high level communities and correspond to research divisions of the institute



Citation networks

- Citations between scientific journals
- Communities represent different scientific disciplines



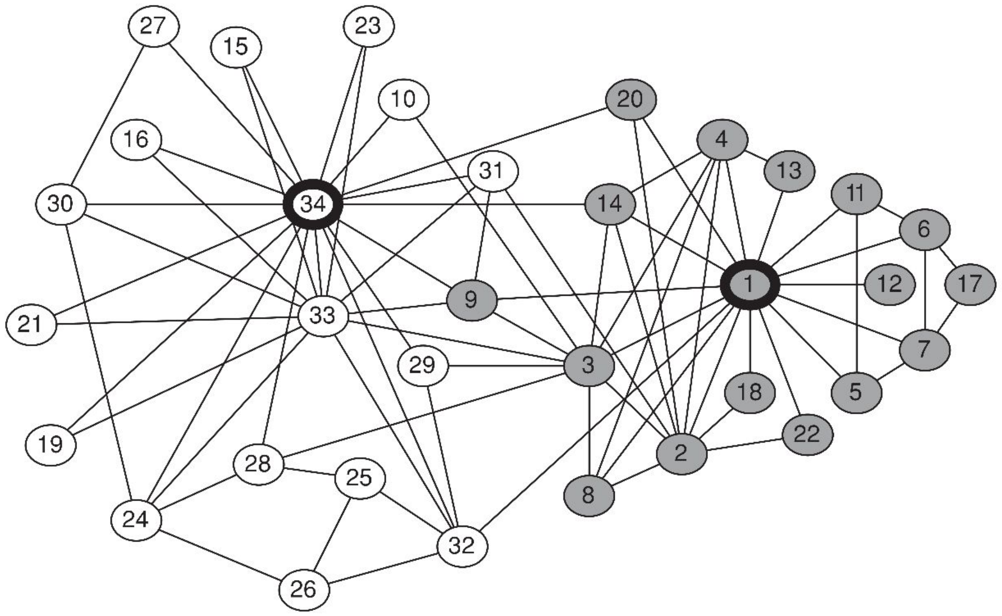
What is community detection?

- The process of finding clusters of nodes ("*communities*")
  - With **Strong** internal connections and
  - **Weak** connections between different communities
- Ideal decomposition of a large graph
  - Completely disjoint communities
  - There are no interactions between different communities.
- In practice,
  - find community partitions that are maximally decoupled.

Why Detecting Communities is Important?

Karate club network

Friendship network between 34 members of a karate club over two years (Zachary, 1972)



- Disagreement between the administrator of the club (node **34**) and the club's instructor (node **1**),
- Led the club to split into two groups (gray and white)
- The members of one group left to start their own club

Communities in the friendship network predicted how the group would split

Community Detection vs. Clustering

Clustering

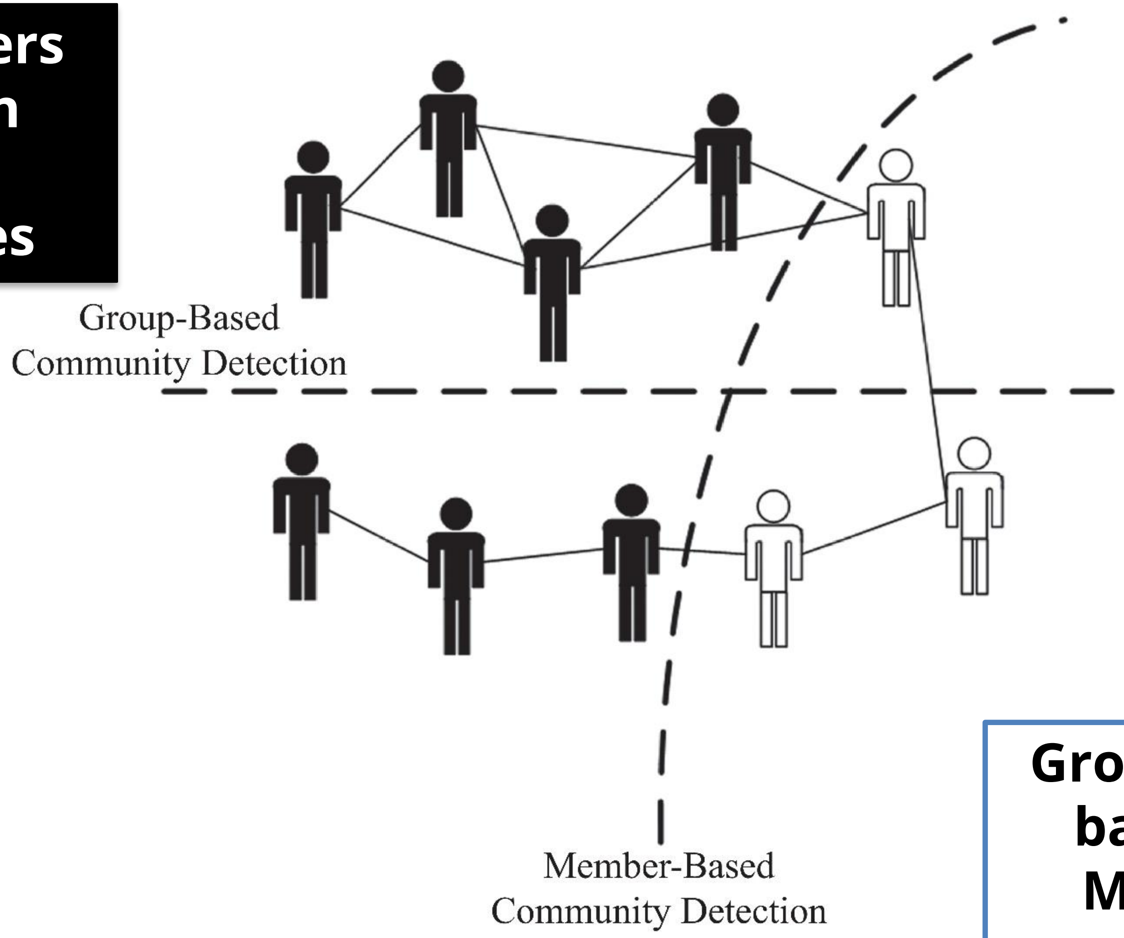
- Data is often non-linked (matrix rows)
- Clustering works on the distance or similarity matrix, e.g., k-means.
- If you use k-means with adjacency matrix rows, you are only considering the ego-centric network

Community detection

- Data is linked (a graph)
- Network data tends to be "discrete", leading to algorithms using the graph property directly
  - k-clique, quasi-clique, or edge-betweenness

Community Detection Algorithms

Group Users based on Group attributes



Group Users based on Member attributes

Member-Based Community Detection

- Look at node characteristics; and
- Identify nodes with similar characteristics and consider them a community

Node Characteristics

A. Degree

- Nodes with same (or similar) degrees are in one community
- Example: cliques

B. Reachability

- Nodes that are close (small shortest paths) are in one community
- Example: k-cliques, k-clubs, and k-clans

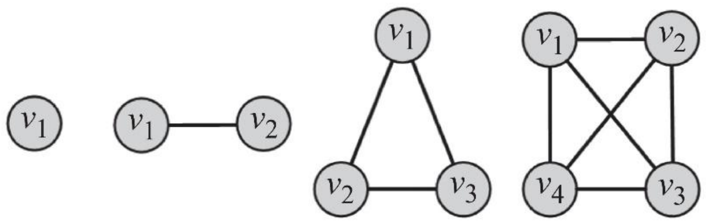
C. Similarity

- Similar nodes are in the same community

A. Node Degree

Most common subgraph searched for:

- **Clique**: a maximum complete subgraph in which all nodes inside the subgraph adjacent to each other



Find communities by searching for

1. **The maximum clique**: **the one** with the largest number of vertices, or
2. **All maximal cliques**: cliques that are not subgraphs of a larger clique; i.e., cannot be further expanded

To overcome this, we can

I. Brute Force
II. Relax cliques
III. Use cliques as the core for larger communities

Both problems are NP-hard

The two extremes

Nodes are assumed to be in the same community

1. If there is a path between them (regardless of the distance) or
2. They are so close as to be immediate neighbors.

How? Find using BFS/DFS

Challenge: most nodes are in one community (giant component)

How? Finding Cliques

Challenge: Cliques are challenging to find and are rarely observed

Solution: find communities that are in between cliques and connected components in terms of connectivity and have small shortest paths between their nodes

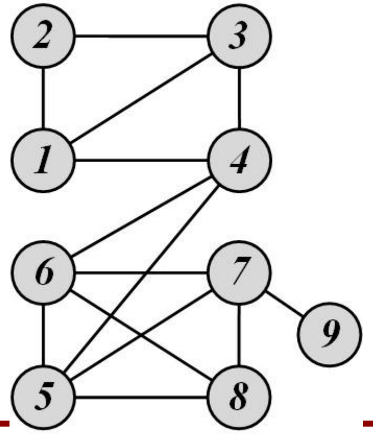
C. Node Similarity

- Similar (or most similar) nodes are assumed to be in the same community.
  - A classical clustering algorithm (e.g., k-means) is applied to node similarities to find communities.

- Node similarity can be defined
  - Using the similarity of node neighborhoods (**Structural Equivalence**)
  - Similarity of social circles (**Regular Equivalence**)

Structural equivalence: two nodes are structurally equivalent iff. they are connecting to the same set of actors

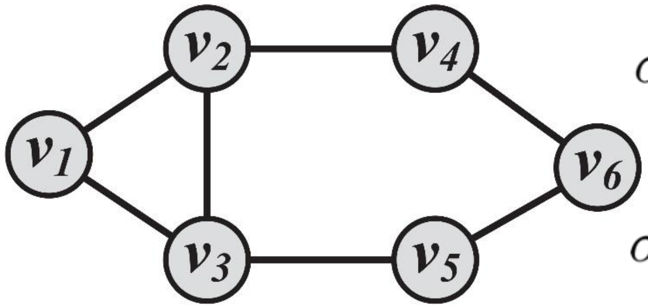
Nodes 1 and 3 are structurally equivalent, So are nodes 5 and 7.



Node Similarity (Structural Equivalence)

Jaccard Similarity $$\sigma_{\text{Jaccard}}(v_i, v_j) = \frac{|N(v_i) \cap N(v_j)|}{|N(v_i) \cup N(v_j)|}$$

Cosine similarity $$\sigma_{\text{Cosine}}(v_i, v_j) = \frac{|N(v_i) \cap N(v_j)|}{\sqrt{|N(v_i)||N(v_j)|}}$$



$$\sigma_{\text{Jaccard}}(v_2, v_5) = \frac{|\{v_1, v_3, v_4\} \cap \{v_3, v_6\}|}{|\{v_1, v_3, v_4, v_6\}|} = 0.25$$

$$\sigma_{\text{Cosine}}(v_2, v_5) = \frac{|\{v_1, v_3, v_4\} \cap \{v_3, v_6\}|}{\sqrt{|\{v_1, v_3, v_4\}||\{v_3, v_6\}|}} = 0.40$$

Group-Based Community Detection

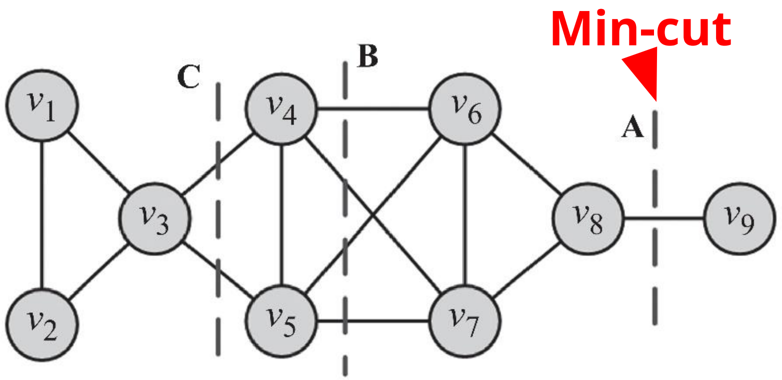
- Group-based community detection: finding communities that have certain **group properties**

Group Properties:

I. **Balanced:** Spectral clustering
II. **Robust:** k-connected graphs
III. **Modular:** Modularity Maximization
IV. **Dense:** Quasi-cliques
V. **Hierarchical:** Hierarchical clustering

I. Balanced Communities

- Community detection can be thought of *graph clustering*
- **Graph clustering:** we cut the graph into several partitions and assume these partitions represent communities
- **Cut**: partitioning (*cut*) of the graph into two (or more) sets (*cutsets*)
  - **The size of the cut** is the number of edges that are being cut
- **Minimum cut (min-cut) problem**: find a graph partition such that the number of edges between the two sets is minimized



Min-cut often returns an imbalanced partition, with one set being a singleton

Ratio Cut and Normalized Cut

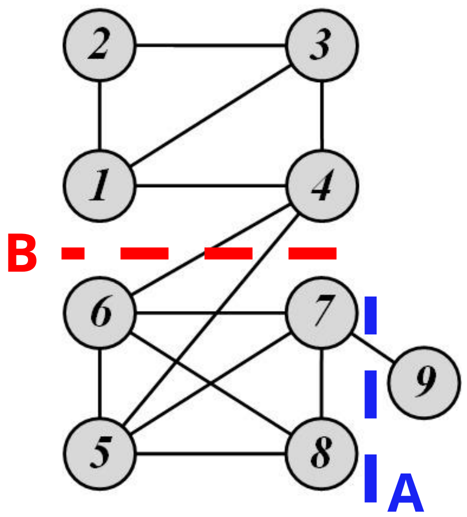
- To mitigate the min-cut problem we can change the objective function to consider community size

$$\text{Ratio Cut}(P) = \frac{1}{k}\sum_{i=1}^{k} \frac{\text{cut}(P_i, \bar{P}_i)}{|P_i|}$$

$$\text{Normalized Cut}(P) = \frac{1}{k}\sum_{i=1}^{k} \frac{\text{cut}(P_i, \bar{P}_i)}{\text{vol}(P_i)}$$

- $\bar{P}_i = V - P_i$ is the complement cut set
- $cut(P_i, \bar{P}_i)$ is the size of the cut
- $vol(P_i) = \sum_{v \in P_i} d_v$

Ratio Cut & Normalized Cut: Example



For Cut A

$$\text{Ratio Cut}(\{1,2,3,4,5,6,7,8\},\{9\}) = \frac{1}{2}\left(\frac{1}{1}+\frac{1}{8}\right) = 9/16 = 0.56$$

$$\text{Normalized Cut}(\{1,2,3,4,5,6,7,8\},\{9\}) = \frac{1}{2}\left(\frac{1}{1}+\frac{1}{27}\right) = 14/27 = 0.52$$

For Cut B

$$\text{Ratio Cut}(\{1,2,3,4\},\{5,6,7,8,9\}) = \frac{1}{2}\left(\frac{2}{4}+\frac{2}{5}\right) = 9/20 = 0.45 < 0.56$$

$$\text{Normalized Cut}(\{1,2,3,4\},\{5,6,7,8,9\}) = \frac{1}{2}\left(\frac{2}{12}+\frac{2}{16}\right) = 7/48 = 0.15 < 0.52$$

Both ratio cut and normalized cut prefer a balanced partition

II. Modular Communities

- Consider a graph $G(V, E)$, where the degrees are known beforehand however edges are not
  - Consider two vertices v_i and v_j with degrees d_i and d_j.

What is an expected number of edges between v_i and v_j?

- For any edge going out of v_i randomly the probability of this edge getting connected to vertex v_j is

$$\frac{d_j}{2m}$$

Modularity and Modularity Maximization

- Given a degree distribution, we know the expected number of edges between any pairs of vertices

- We assume that real-world networks should be far from random. Therefore, the more distant they are from this randomly generated network, the more structural they are.

- Modularity defines this distance and modularity maximization tries to maximize this distance

Normalized Modularity

- Consider a partitioning of the graph $P = (P_1, P_2, P_3, \ldots, P_k)$

For partition P_x, this distance can be defined as

$$\sum_{i,j \in P_x} A_{ij} - \frac{d_i d_j}{2m}$$

This distance can be generalized for a partitioning P

$$\sum_{x=1}^{k} \sum_{i,j \in P_x} A_{ij} - \frac{d_i d_j}{2m}$$

The normalized version of this distance is defined as Modularity

$$Q = \frac{1}{2m} \sum_{x=1}^{k} \sum_{i,j \in P_x} A_{ij} - \frac{d_i d_j}{2m}$$

Recall:
Measuring Assortativity for Nominal Attributes

- Assume **nominal** attributes are assigned to nodes
  - Example: race
- Edges between nodes of the same type can be used to measure assortativity of the network
  - Same type = nodes that share an attribute value
  - Node attributes could be nationality, race, sex, etc.

$$\frac{1}{m}\sum_{(v_i,v_j)\in E}\delta(t(v_i),t(v_j)) = \frac{1}{2m}\sum_{ij}A_{ij}\delta(t(v_i),t(v_j))$$

$t(v_i)$ denotes type of vertex v_i

$$\delta(x,y) = \begin{cases} 0, & \text{if } x \neq y \\ 1, & \text{if } x = y \end{cases}$$

Kronecker delta function

Assortativity Significance

- **Assortativity significance**
  - The difference between **<u>measured assortativity</u>** and **<u>expected assortativity</u>**
  - The higher this difference, the more significant the assortativity observed

Example

- In a school, 50% of the population is **white** and the other 50% is **hispanic**.
- We expect 50% of the connections to be between members of different races.
- If all connections are between members of different races, then we have a significant finding

Assortativity Significance

Assortativity

Expected assortativity

$$Q = \boxed{\frac{1}{2m}\sum_{ij} A_{ij}\delta(t(v_i), t(v_j))} - \boxed{\frac{1}{2m}\sum_{ij} \frac{d_i d_j}{2m}\delta(t(v_i), t(v_j))}$$

This is modularity

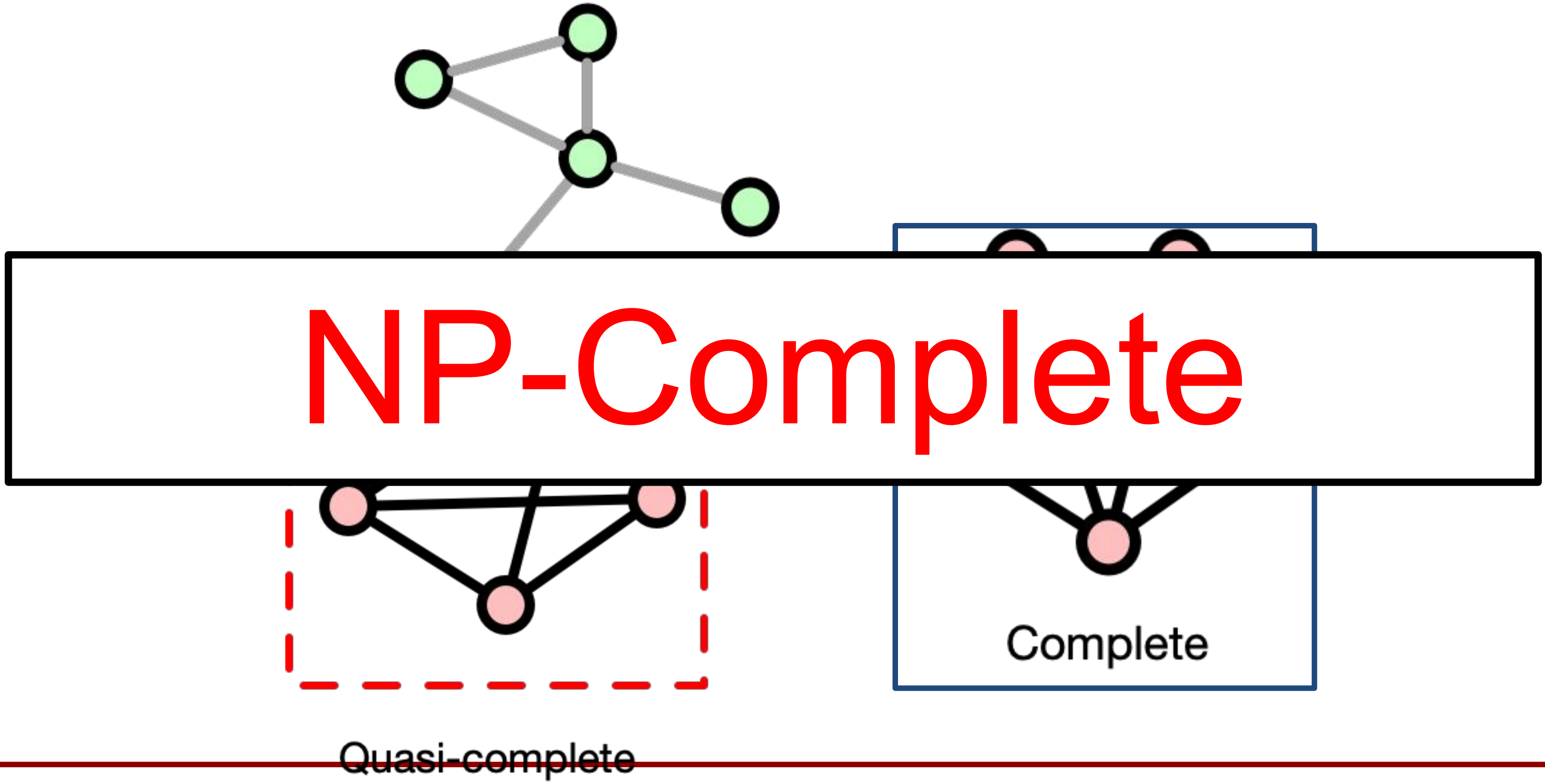
How do we maximize modularity?

- Intuition: maximizing Q is hard
- Imagine communities as dense subgraph
- Simplest dense subgraph: all-to-all (cliques)
- This is NP-hard

NP-Complete

Simplifying the problem

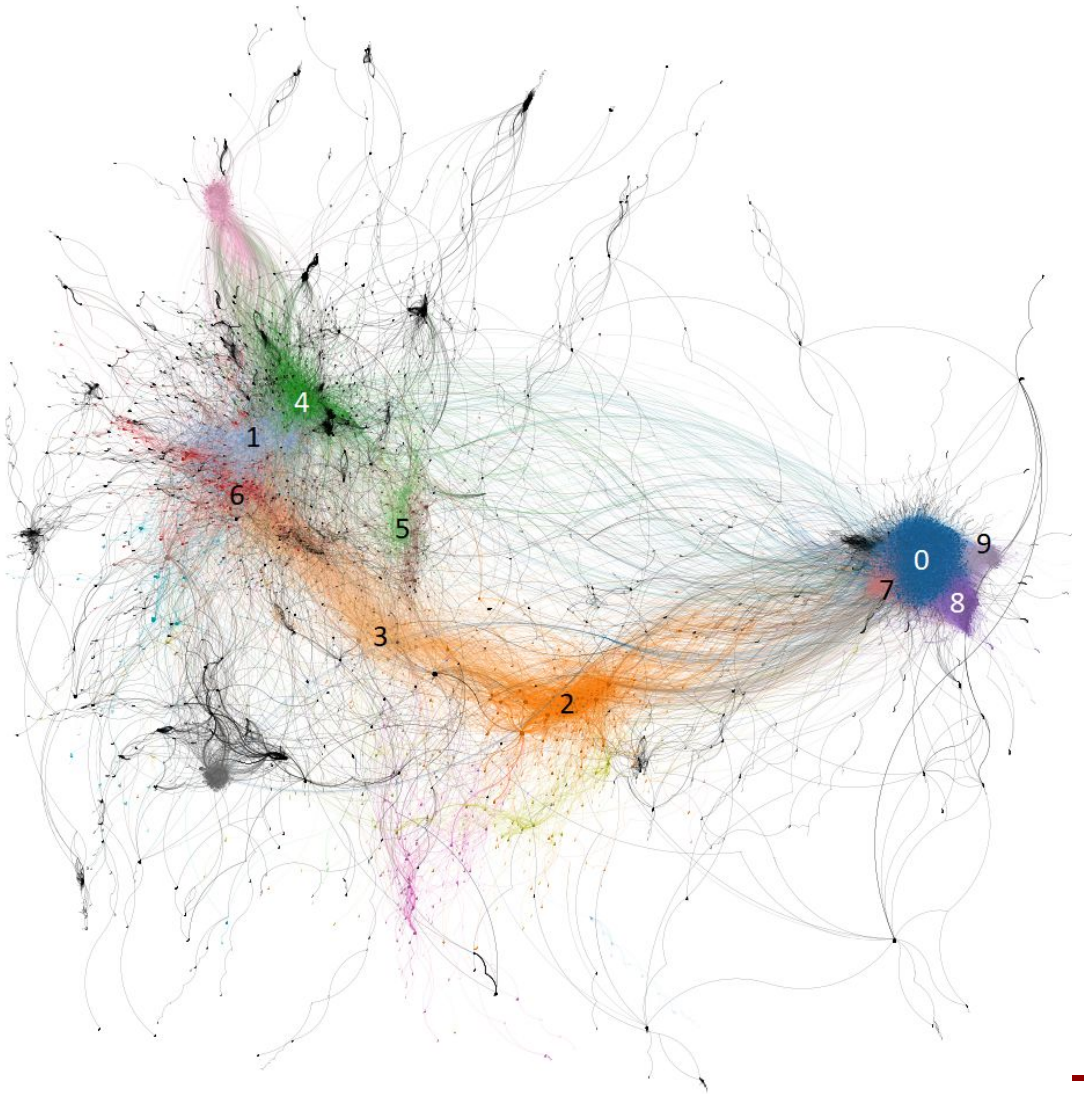
What about a quasi-complete subgraph? i.e., almost-all-to-all subgraph?



Modularity Optimization

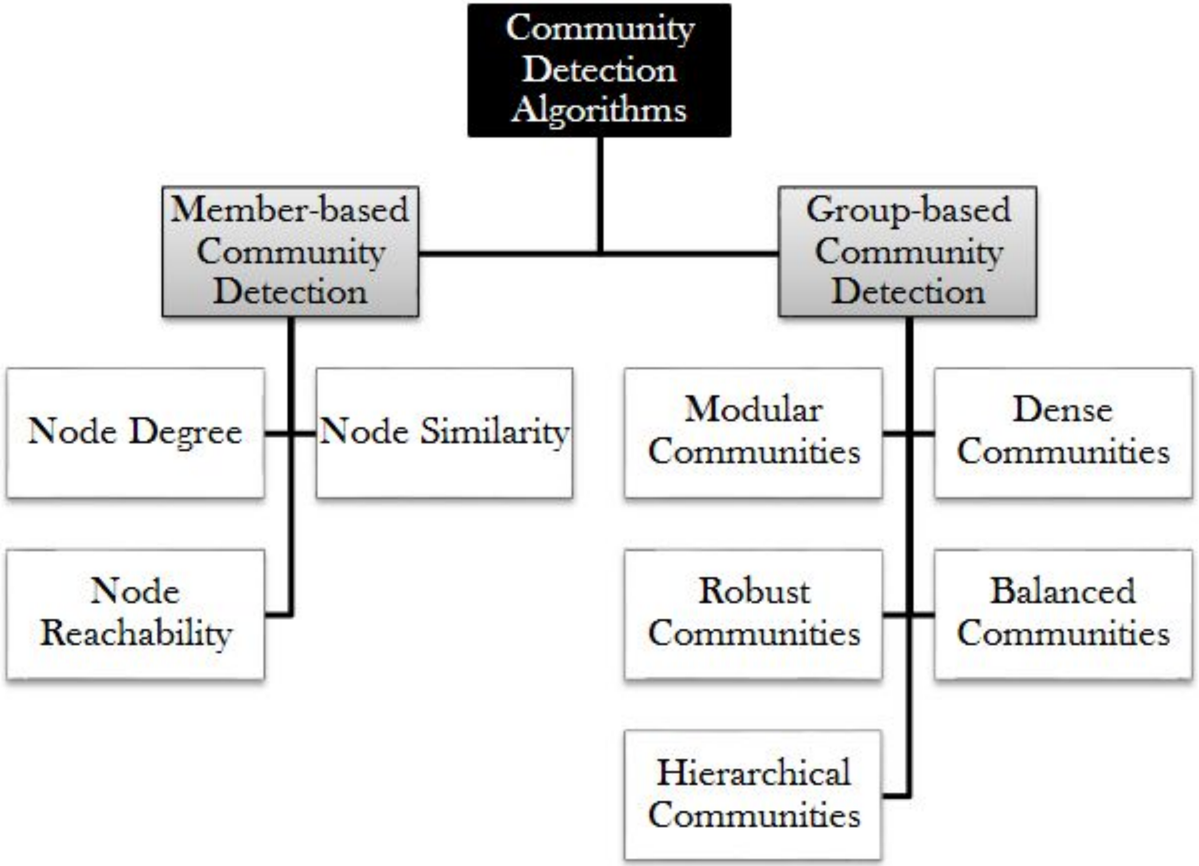
- Assume greedy algorithm is good enough
- Louvain Method
  - A greedy modularity optimization method for community detection
    - Invented when all authors affiliated with ***Université catholique de Louvain (UCL)***

Example: communities in a retweet network



- Retweet network between users
- Communities (distinct colors) identified by Louvain modularity
- Analysis of tweets within communities reveals coherent topics of discussion within each community

Community Detection Algorithms

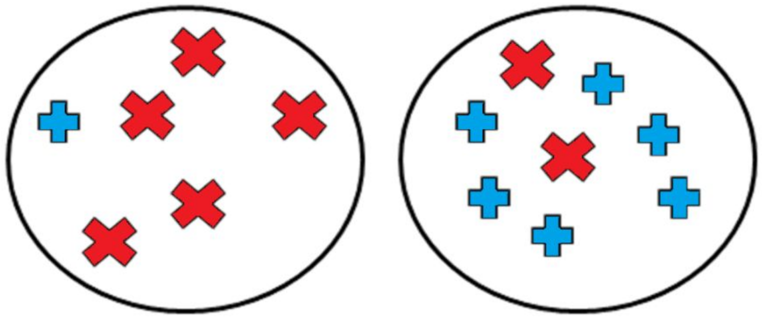


COMMUNITY EVALUATION

Evaluating the Communities

We are given objects of two different kinds (+, ×)

- **The perfect community:** all objects inside the community are of the same type



- **Evaluation with ground truth**
- **Evaluation without ground truth**

Evaluation with Ground Truth

- When ground truth is available
  - We have partial knowledge of what communities should look like
  - We are given the correct community (clustering) assignments

- **Measures**
  - Precision and Recall, or F-Measure
  - Purity
  - Mutual Information

Purity

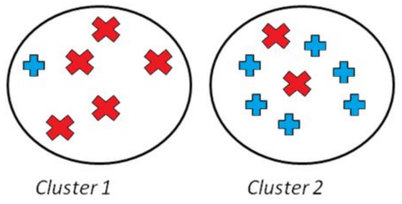
We can assume the majority of a community represents the community

– We use the label of the majority against the label of

Purity can be easily **tampered** by

- Points being singleton communities (of size 1); or by
- Very large communities

$$Purity = \frac{1}{N}\sum_{i=1}^{k}\max_j |C_i \cap L_j|$$

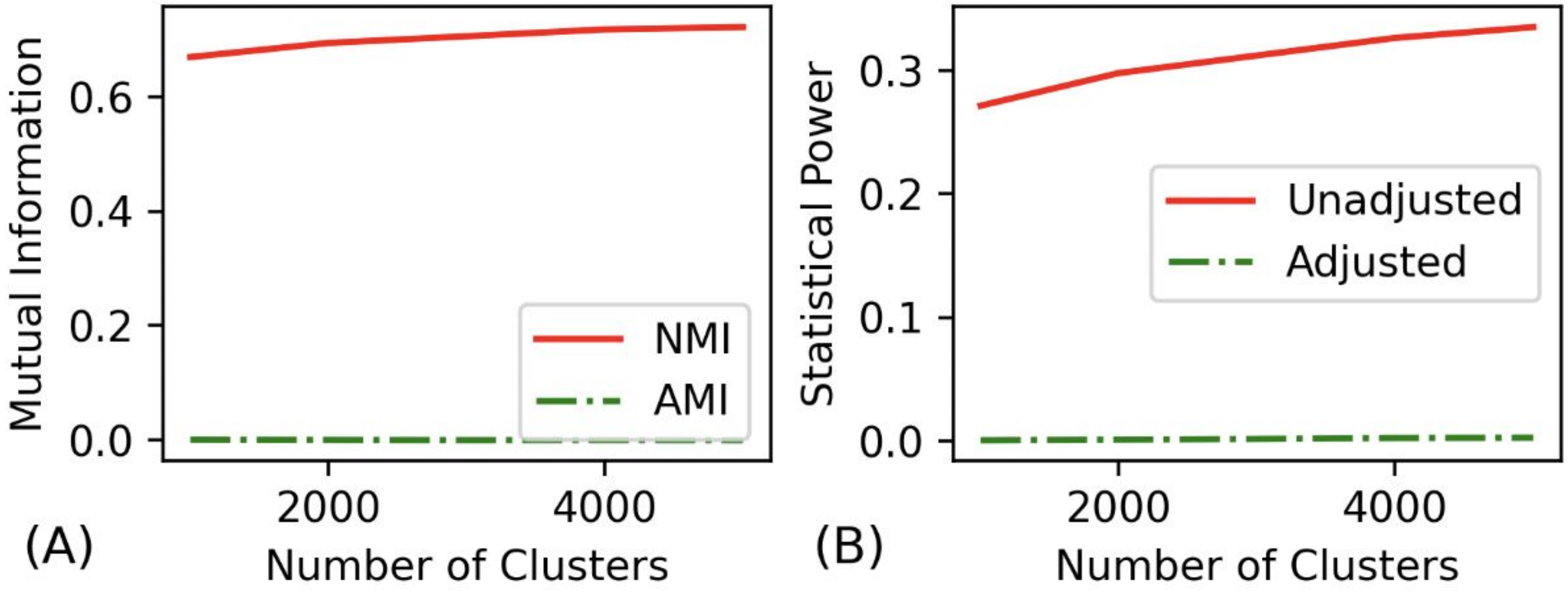


$purity\ is:\ \frac{6+5}{14} = 0.78$

- k: the number of communities
- N: total number of nodes,
- L_j: the set of instances with label j in all communities
- C_i: the set of members in community i

Mutual information or statistical power

- Other options include (normalized) mutual information or statistical power
- These metrics increase with more (false) clusters (Adjusted MI is better!)



Evaluation without Ground Truth

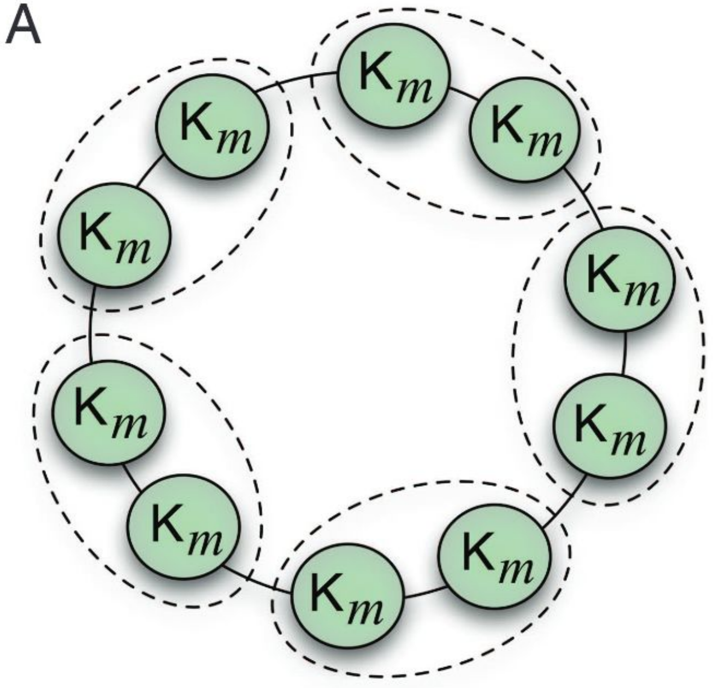
- **Evaluation with Semantics**
  - A simple way of analyzing detected communities is to analyze other attributes (posts, profile information, content generated, etc.) of community members to see if there is a coherency among community members
  - The coherency is often checked via human subjects.
  - To help analyze these communities, one can use word frequencies. By generating a list of frequent keywords for each community, human subjects determine whether these keywords represent a coherent topic.

- **Evaluation Using Clustering Quality Measures**
  - Use clustering quality measures (SSE)
  - Use more than two community detection algorithms and compare the results and pick the algorithm with better quality measure

Evaluate with Modularity?

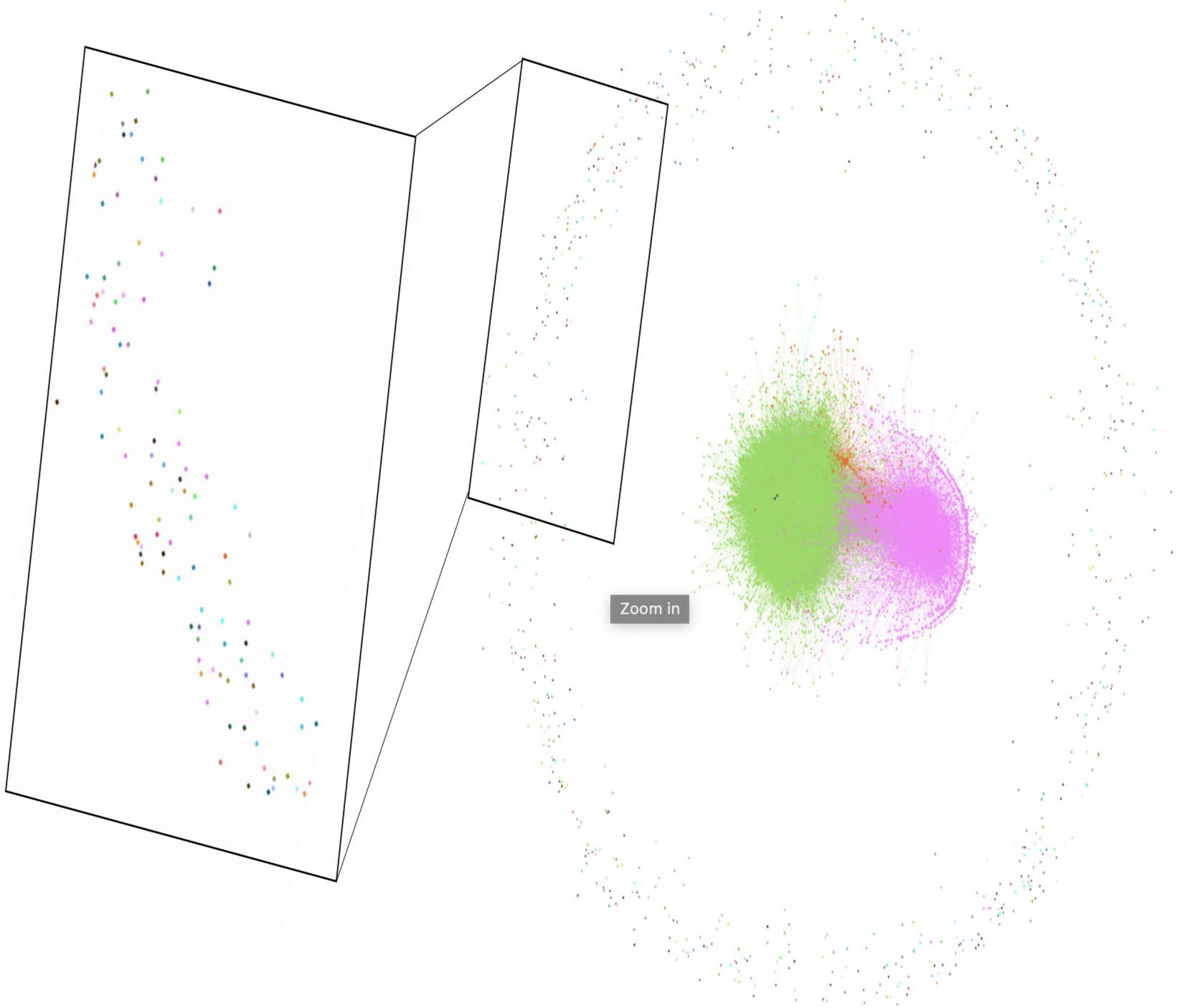
- Often modularity used to evaluate without GT
- (May be bad): resolution limit
  - Create graph with all-to-all subgraphs
  - Connect subgraphs with 1 link
  - If cliques > sqrt(|E|) (more than square root of the number of edges): combining cliques increases modularity
  - This means modularity ignores cliques that are too small

A



Bias in community detection

- Traditional methods **fail to assign low-degree (lowly connected) users to significant communities**.
- These users are either ignored or placed into **small, insignificant groups**, leading to **biased community analysis**.
- This omission causes the loss of **relevant information** from these users, despite their potential significance to the network's discussions.



Mehrabi, et al (2019). Debiasing community detection: the importance of lowly connected nodes. In *Proceedings of ASONAM* (pp. 509-512).

Bias in community detection

- The study examined two datasets:
  - **Gamergate Twitter dataset** (2014)
  - **2016 U.S. Presidential Election Twitter dataset**
- They showed that significant hashtags, such as **#refugeeswelcome, #Christians4Hillary, #gamersagainstgamergate**, were excluded from major community analyses because the users who posted them were placed in insignificant groups.

TABLE III
THE QUANTITATIVE RESULTS OBTAINED FROM CALCULATING THE F1 AND JACCARD SIMILARITY SCORES WITH REGARDS TO THE GROUND TRUTH LABELS FOR EACH OF THE METHODS ALONG WITH THE NUMBER OF SIGNIFICANTLY LABELED USERS.

Method	Gamergate			U.S. Election		
	Significantly Labeled	F1 Score	Jaccard	Significantly Labeled	F1 Score	Jaccard
CESNA	0.31	0.343	0.211	0.05	0.253	0.149
Modularity	0.79	0.434	0.282	0.80	0.753	0.604
CLAN	**1.00**	**0.478**	**0.318**	**1.00**	**0.787**	**0.649**